

01_HARDWARE_ARCHITECTURE

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EVIDENCE PROVENANCE: [PHYSICAL HARDWARE]

Direct inspection and measurement of dual ESP32 prototype chassis, optical tachometers, IMU, and motor drivers.

1. Prototype Specifications Matrix

Component	Vehicle A (TRUCK_01)	Vehicle B (TRUCK_02)	Notes
Microcontroller	ESP32-WROOM-32	ESP32-WROOM-32	Dual-core 240 MHz, 520KB SRAM
Motor Driver	L298N Dual H-Bridge	Toshiba TB6612FNG	Independent H-bridge architectures
Optical Encoder	Slotted IR (LM393)	Slotted IR (LM393)	GPIO 35 interrupt, RISING edge
Physical Slots (PPR)	42 slots	43 slots	Physical disc count difference
Calibration K_cal	34.58 pulses/rev	34.58 pulses/rev	Empirical divisor under loaded bench
Wheel Diameter	0.060 m (60 mm)	0.060 m (60 mm)	Outer tread diameter (C = 0.1885 m)
6-DOF IMU	MPU6050 (I2C 0x68)	MPU6050 (I2C 0x68)	SDA: 21, SCL: 22
LoRa Transceiver	Semtech SX1278 (433MHz)	Semtech SX1278 (433MHz)	SPI bus (18, 19, 23, 5, 4, 2)
CAN / TWAI Bus	250 kbps SAE J1939	250 kbps SAE J1939	Commercial vehicle protocol parity

2. Odometry Calculation Equations

- Wheel Circumference:** $C = \pi * D = 0.188496 \text{ m}$.
- Effective Distance Resolution:** $\text{delta}_s = C / K_{cal} = 0.188496 / 34.58 = 5.451 \text{ mm/pulse}$.
- Velocity Equation:** $v = (\text{pps} * C) / 34.58 \text{ m/s}$ (calculated at 10 Hz interval).